Myoelectric Control Based on a Generic Musculoskeletal Model: Toward a Multi-User Neural-Machine Interface

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***Abstract*— This paper aimed to develop a novel electromyography (EMG)-based neural-machine interface (NMI) that is user-generic for continuously predicting coordinated motion between metacarpophalangeal (MCP) and wrist flexion/extension. The NMI requires a minimum calibration procedure that only involves capturing maximal voluntary muscle contraction for the monitored muscles for individual users. At the center of the NMI is a user-generic musculoskeletal model based on the experimental data collected from six able-bodied (AB) subjects and nine different upper limb postures. The generic model was evaluated on-line on both AB subjects and a transradial amputee. The subjects were instructed to perform a virtual hand/wrist posture matching task with different upper limb postures. The on-line performance of the generic model was also compared with that of the musculoskeletal model customized to each individual user (called "specific model"). All subjects accomplished the assigned virtual tasks while using the user-generic NMI, although the AB subjects produced better performance than the amputee subject. Interestingly, compared with the specific model, the generic model produced comparable completion time, a reduced number of overshoots, and improved path efficiency in the virtual hand/wrist posture matching task. The results suggested that it is possible to design an EMG-driven NMI based on a musculoskeletal model that could fit multiple users, including upper limb amputees, for predicting coordinated MCP and wrist motion. The present new method might address the challenges of existing advanced EMG-based NMI that require frequent and lengthy customization and calibration. Our future research will focus on evaluating the developed NMI for powered prosthetic arms.**

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I. Introduction

RECENTLY, robotic prostheses with multiple electric joints have become commercially available. However, the dominant myoelectric control scheme for these robotic prostheses is still conventional direct control, in which the amplitude of EMG signals from an agonist-antagonist muscle pair is used to control one degree of freedom (DoF). One limitation of this control scheme is that a non-intuitive switching command is required for sequential control of two or more DoFs, which may explain why only one-fourth of upper limb amputees use prostheses with direct control in daily life [1], [2].

EMG decoders based on advanced data-driven algorithms, which explore data structures and patterns based on the data (inputs and outputs) without any knowledge of the measured system, have been proposed for multifunctional prosthesis control for decades [3]–[13]. Even within the last decade, researchers have been intensively exploring various advanced data-driven, machine learning approaches (such as adaptive classifiers, neural networks, regression/filtering methods, etc.) to decipher EMG signals for multifunction prosthesis control [7], [8], [10]–[14]. The data-driven approaches usually depend on the training data, collected during the calibration procedure. In this procedure, the user needs to perform instructed hand/arm motions repetitively (usually 3-5 times per motion), while EMG signals are collected. Then, the collected data (i.e. training data) consists of known input EMG signals and output user intended motions (i.e. instructed hand/arm motions). The training data is used to establish boundaries of data pattern classifiers or input-output data mapping relationship for real-time use. The calibration procedure usually takes 5-10 minutes, depending on the number of repetitions for collecting the training data [15], [16].

One of the challenges for the data-driven EMG decoders is the lack of ability for learning transfer. That is to say, the data-driven algorithm, trained by one EMG set, might not be generalizable to new data sets that present different or varied information from the training data. For example, due to inter-human variation, EMG-based NMI trained by data collected

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from one person usually cannot be transferred to the data collected from another. Therefore, the NMI must be customized to individual users. The lack of learning transfer is also demonstrated by the evidence that the performance of the data-driven EMG NMI decreases over time [8], [17]. This is because the NMI is not generalized enough to handle EMG recording variations caused by electrode location shift, electrode-skin impedance change, muscle fatigue, arm posture changes, etc. over daily use [17]. Hence, recalibration is needed at least once a day and potentially multiple times during daily use. Such frequent recalibration potentially diminishes the practicality of these advance EMG-based NMIs. Researchers have tried to address this problem with exploring different training strategies [18], resilience EMG features [19] and adaptive classifiers [8], [17], [20], [21]. Prostheses-guided training has been proposed to reduce the training time to about 1 minute and result in functional systems that perform equivalently or better than direct control systems in home-trials with highly variable and subject-specific recalibration habits [22], [23]. These methods showed the promise for reducing the number of recalibrations in daily use for each individual user. Some generalized NMI have been developed for control across users without recalibration. For example, one generalized NMI system based on information fusion of accelerometer and EMG sensors was demonstrated for hand language gesture recognition on limb-intact subjects [24]. Another generalized NMI system based on EMG towards gesture recognition on able-bodied subjects was reported but demonstrated a relatively low classification accuracy of 57% [25]. However, a generalized NMI based on EMG towards neural rehabilitation that can be used across human users, to our knowledge, has not been demonstrated.

The data-driven approach tries to find relationship between EMG signals (neuromuscular control signals) and upper limb joint motions without considering the biological process between the two measurements. This might explain the lack of ability for learning the generalized relationship between the input and output in data-driven EMG NMI. Recently several research groups, including our group, started to explore the possibility of building EMG NMI based on EMG-driven musculoskeletal models [26]–[28]. Specially in our group, an EMG NMI based on a simplified musculoskeletal model was developed for predicting simultaneous, continuous flexion/extension movements at the metacarpophalangeal (MCP) and wrist joints [27], [29], [30]. The EMG interface directly incorporates the physiological and biomechanical structures and constraints of the neuromusculoskeletal system of the hand. In our initial design, we built an EMG-driven musculoskeletal model customized for individual users. We found that the model-based NMI was inherently reliable across different use conditions. Two factors might contribute to the reliability of this EMG decoder: (1) the primary biomechanical actions of the muscles crossing the wrist and fingers are likely preserved across limb postures, and (2) humans can adapt to the model-based EMG interface and compensate for signal variations caused by posture change.

Given the consistent performance of the model-based EMG decoder customized for each subject, in this study we questioned whether such reliability could be extended across subjects. Perhaps due to the aforementioned two factors, customizing the musculoskeletal model is not necessary. Rather, one generic model could be used for any users after normalizing EMG inputs by maximal voluntary muscle contraction (MVC). This idea is intriguing because, if a generic EMG-driven model could be used for multifunctional prosthesis control, it would significantly simplify the calibration procedure and enable natural prosthesis control at the same time.

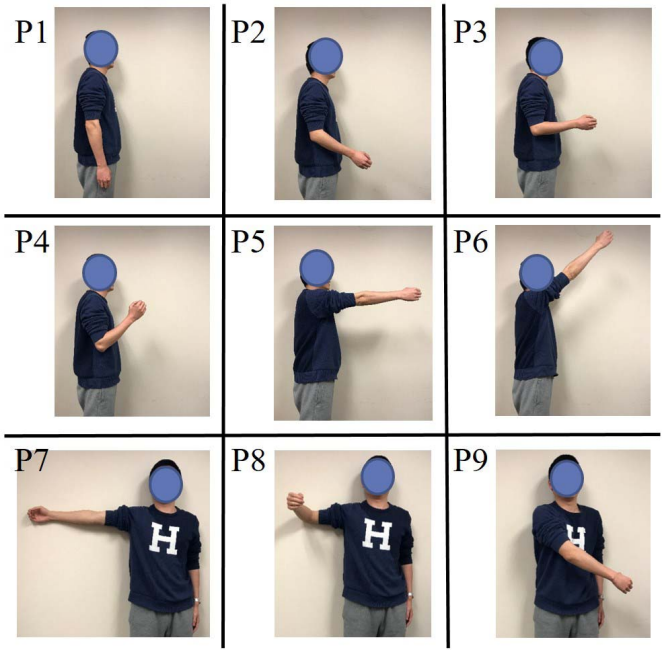

Fig. 1. The 9 different postures tested in the experiments.

II. Methods

A. Subjects

The experimental protocol was approved by the University of North Carolina at Chapel Hill Institutional Review Board. Eleven subjects, including ten able-bodied (AB) subjects and one transradial amputee, were recruited in this study. Six AB subjects (all male, ages 19-32, right hand dominant) participated in the off-line experiment. Seven AB subjects, including three AB subjects who also participated in the off-line experiment, (5 male, ages 25-32, right hand dominant; 2 female, ages 26-30, left hand dominant) and one transradial amputee subject (female, age 23, right mid-forearm amputation) participated in the on-line experiment. The subjects provided informed consent to participate. The amputee subject used a direct control 1-DoF myoelectric prosthesis for daily activities.

B. Off-Line Experimental Design and Data Collection

1) Data Acquisition and Processing: EMG and kinematic data were collected synchronously from six AB subjects in nine upper limb postures (P1-P9, Fig. 1). In each posture, all subjects performed 5 different movements with either fixed speed (rhythm) or variable speed (random): (1) metacarpophalangeal (MCP) flexion/extension only, rhythm; (2) wrist

Fig. 2. Experimental setup for off-line data collection. The positions of the markers and the surface EMG electrodes on an AB subject.

flexion/extension (WFLX/EXT) only, rhythm; (3) MCP flexion/extension only, random; (4) WFLX/EXT only, random; (5) simultaneous MCP flexion/extension and WFLX/EXT, random.

For kinematic data acquisition, the three-dimensional positions of 14 retro-reflective markers were collected at 120 Hz by an infrared motion capture system (Vicon Motion Systems Ltd. UK). The markers were mounted on the following anatomical locations (Fig. 2): 2nd and 5th proximal interphalangeal joint (PIP) joints, 2nd and 5th MCP joints, radial styloid process, ulnar styloid process, radial midshaft, medial humeral epicondyle, lateral humeral epicondyle, humeral midshaft, medial clavicle, acromion, C7 spinous process, and xiphoid. Since the joint angles of the hand and wrist could not be directly measured from the amputee's residual limb, the 14 retro-reflective markers were mounted on the intact side to acquire kinematic data during the mirrored bilateral movements; additional markers were placed on the residual limb to monitor its posture during the test.

Four-channel surface EMG signals were acquired from four extrinsic hand- and wrist-related muscles (Fig. 2): extensor carpi radialis longus (ECRL), extensor digitorum (ED), flexor carpi radialis (FCR), and flexor digitorum (FD). The EMG signals were high-pass filtered (4th-order Butterworth filter with a cutoff frequency at 40 Hz), rectified, low-pass filtered (4th-order Butterworth filter with a cutoff frequency at 6 Hz), and then normalized by the maximum EMG values recorded during maximum voluntary contraction (MVC). The normalized EMG signals were down-sampled to 120 Hz and used to compute muscle activations using a non-linear model of excitation-activation dynamics [31].

2) Muscle Modeling and Parameters Optimization: In our research group, a lumped-parameter forward musculoskeletal model has been developed as an EMG-based NMI [27]. The

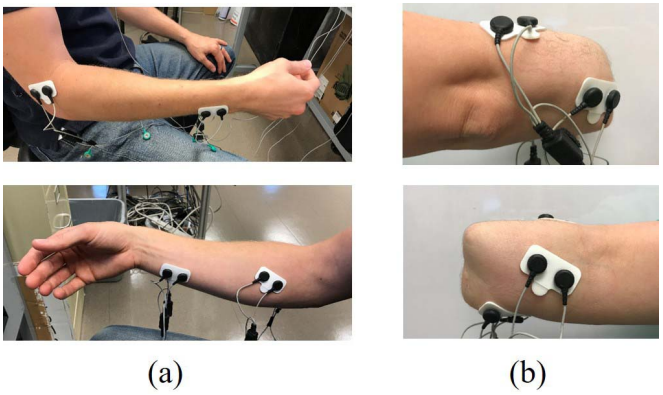

Fig. 3. Experimental setup for on-line testing. (a) The positions of the surface EMG electrodes on an AB subject; (b) The positions of the surface EMG electrodes on the amputee subject.

modeling approach used in this paper was more fully defined and described in the previous paper [27]. Each muscle was modeled as a Hill-type actuator with a contractile element and a parallel elastic element [32]. To reduce the number of model parameters, a series elastic element representing tendon in the Hill-type model was not included. Six parameters were optimized for each muscle: optimal muscle length, maximum isometric force, moment arms at the MCP and wrist joints, parallel elastic element stiffness and muscle length at neutral position. Moment arms were set to zero at joints which a muscle does not cross anatomically. All parameters were constrained to approximate ranges of physiologic values [27]. All moment arms were assumed constant across joint angles.

The MATLAB (Mathworks, Inc, MA) optimization function GlobalSearch was used to compute the muscle parameters in each of the nine postures for each of the six AB subjects. That is to say, we used a portion of the data collected from one posture to establish the parameters in one musculoskeletal model. Thus, for each subject, there were 9 musculoskeletal models, one for each posture. The experimentally-derived muscle activations and estimated 2-DoF joint angles were the inputs and outputs of the model, respectively, during a forward dynamics simulation. The muscle parameters were optimized to minimize the mean squared error between estimated and measured joint angles. To develop a generic musculoskeletal model for the later on-line test, we averaged the model parameter values across all subjects and all postures. The parameters of the generic musculoskeletal model have been included with the manuscript as supplementary data.

C. On-Line Testing Protocol

1) Data Acquisition and Processing: Four-channel surface EMG signals were acquired from the same four muscles (Fig. 3) as in the off-line experiments. The skin surface over the muscles was cleaned with alcohol to reduce the impedance. Four bipolar adhesive surface EMG electrodes (Norotrode 20, Myotronics Inc., WA), which had an electrolyte gel to reduce skin-electrode impedance, were mounted over the target muscles. The electrodes were connected to a preamplifier (MA-420, Motion Lab Systems, Inc., LA). The surface

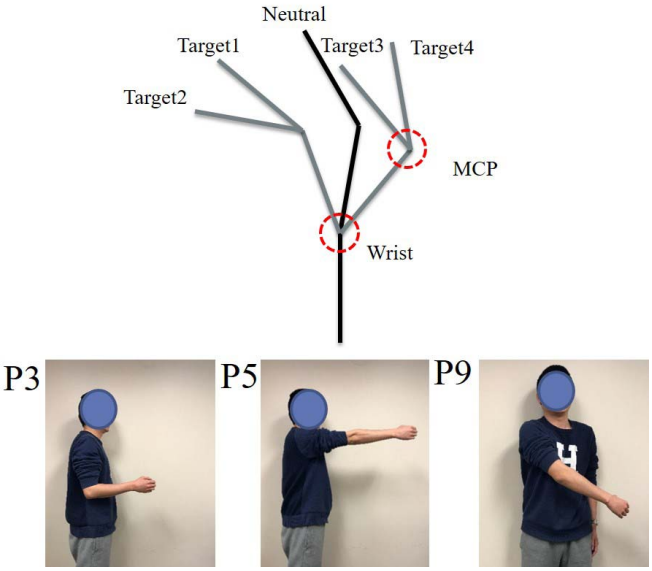

Fig. 4. Diagram of four target hand postures (gray lines) and neutral hand posture (black line).

EMG signals were recorded at 1000 Hz (MA300, Motion Lab Systems, Inc., LA).

The EMG signals were sampled in sequential 100 ms windows and accumulated over the 20 most recent windows. The accumulated EMG were high-pass filtered (1st-order Butterworth zero-lag filter with a cutoff frequency at 40 Hz), rectified, and low-pass filtered (a 1st-order Butterworth zero-lag filter with a cutoff frequency at 1 Hz). EMG from the 2nd most recent 100 ms window were downsampled to 100 Hz by averaging EMG across each 10 consecutive time points, and normalized by the maximum EMG recorded during maximum voluntary contractions (EMG-MVC). Finally, a moving average was computed across the 7 most recent downsampled points to estimate muscle activations of the four virtual muscles at each time point.

2) Model Description: In two separate on-line experiment sessions, we tested either the specific model customized for each subject in upper limb posture P3 in our previous studies [27], [33] or the generic model developed from the above off-line experiments on the AB subjects. The order of these two sessions was randomized. For the amputee subject, we tested the generic model only in one on-line experiment session. The model was implemented for on-line EMG-driven control of a planar stick figure virtual hand in MATLAB [30]. Muscle activations ranging from 0% to 100% were computed from recorded EMG, as described above, and applied to the virtual muscles during an on-line forward dynamics simulation to estimate MCP and wrist joint angles. The posture of the virtual hand, which included the same DOFs as the musculoskeletal model, was displayed on a computer screen and updated at 20 Hz.

3) Pre-Task Model Calibration: After placing EMG electrodes, subjects were seated in front of a monitor and held their upper limb in a static upper limb posture (P3, Fig. 4). EMG-MVC were recorded while subjects sequentially performed maximum voluntary contractions of MCP and wrist flexion and extension. The EMG-MVC recording process took 15 seconds. Subjects then attempted to move the virtual hand to each of the four tested target hand postures (Fig. 4). An experimenter adjusted linear gains manually to scale the input (EMG) and output (joint angles) so that each subject could (1) reach a neutral hand posture when relaxed and (2) reach all targets comfortably. The gain adjustment process took about 1 minute. Once it was done, there was no recalibration needed during the on-line test.

4) On-Line Virtual Task: In each trial, subjects were instructed to move the virtual hand from the neutral hand posture to one of four target hand postures (Fig. 4) as quickly as possible. To ensure that subjects achieved each posture in a controlled manner, subjects were required to hold the virtual hand within ± 8 degrees of the neutral posture for 0.25 consecutive seconds and within ± 5 degrees of the target posture for 0.5 consecutive seconds. The neutral and target postures turned green when the subject was within the acceptable range. In each trial, subjects attempted all target hand postures in a randomized order. All subjects performed one familiarization trial and 4 test trials each with their upper limb in 3 different testing static upper limb postures (P3, P5, and P9, Fig. 4) in a randomized order. We recorded the EMG signals, muscle activations, and virtual hand kinematics for each trial. To quantify the on-line control performance, completion time, number of overshoots, and path efficiency were employed. The completion time was defined as the time used by the subject to accomplish the targets. The number of overshoots was defined as the number of times that the virtual hand moved in and out of the target posture range. The path efficiency was defined as the ratio between the length of the actual path performed by the subject and the length of the optimal path, in joint angle Cartesian space.

D. Statistical Analysis

For the AB subjects, three-way repeated measures analysis of variance (ANOVA) was conducted on completion time, the number of overshoots, and path efficiency. ANOVA included the following independent variables: model type (generic vs. specific), posture (3 different upper limb postures), and target type (4 different targets). In all ANOVA tests, a full statistical model was generated first. When ANOVA revealed a significant interaction, a simple-effects analysis was performed by fixing the levels of one of the interacting factors. If no significant interaction was revealed, a reduced ANOVA model with only main effects was conducted. The significance level for all tests was set to $p = 0.05$.

III. Results

A. On-Line Testing Results

Across AB subjects and trials, the average completion times was around 6 s for all targets in 3 different testing postures for the specific model and the generic model (Fig. 5). The average completion time for the specific model in P3, P5, and P9 was 6.64 s, 6.54 s, and 5.87 s, respectively. The average completion time for the generic model in P3, P5, and P9 was 6.64 s, 6.54 s, and 5.87 s, respectively. For

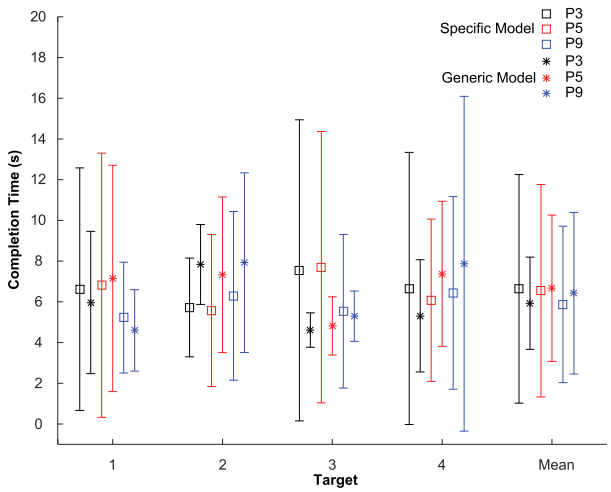

Fig. 5. Average completion time for each target across AB subjects and trials. The rightmost six points are the average values across all targets. Error bars represent the standard deviations.

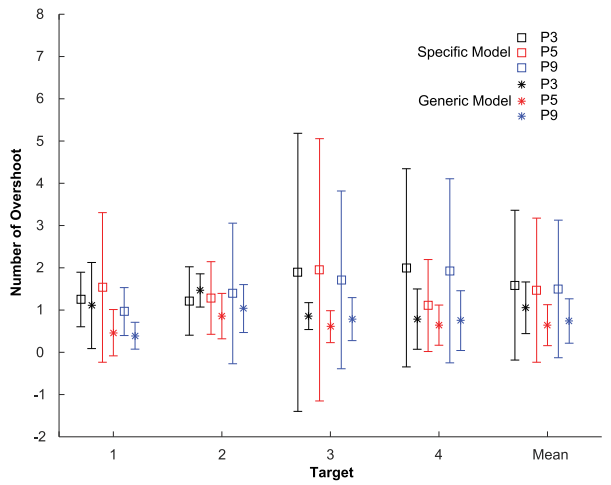

Fig. 6. Average number of overshoots for each target across AB subjects and trials. The rightmost six points are the average values across all targets. Error bars represent the standard deviations.

target 3, the completion times and their standard deviations for the specific model were higher than that for the generic model in the 3 different testing postures, but this difference was not statistically significant. The three-way ANOVA revealed no significant three-way or two-way interactions. ANOVA with only main effects indicated that the model type, posture, and target type had no statistically significant effects (model type: $p = 0.88$, posture: $p = 0.81$, and target type: $p = 0.34$) on completion time.

Fig. 6 reports the average number of overshoots for each target across AB subjects and trials. The number of overshoots was around 1.5 for all targets in the 3 different testing postures for the specific model. The number of overshoots was around 0.8 for all targets in the 3 different testing postures for the generic model. The average number of overshoots for the specific model in P3, P5, and P9 was 1.59, 1.47, and 1.5, respectively. The average number of overshoots for the generic model in P3, P5, and P9 was 1.05, 0.64, and 0.74, respectively. For target 3, the number of overshoots and their standard deviations for the specific model were higher than that for the generic model in the 3 different testing postures. The three-way ANOVA revealed no three-way or two-way interactions. ANOVA with only main effects indicated that the number of overshoots for the generic model (0.81) was significantly lower ($p < 0.001$) than that for the specific model (1.52), while posture and target type had no statistical effects (posture: $p = 0.2$ and target type: $p = 0.45$) on the numbers of overshoots.

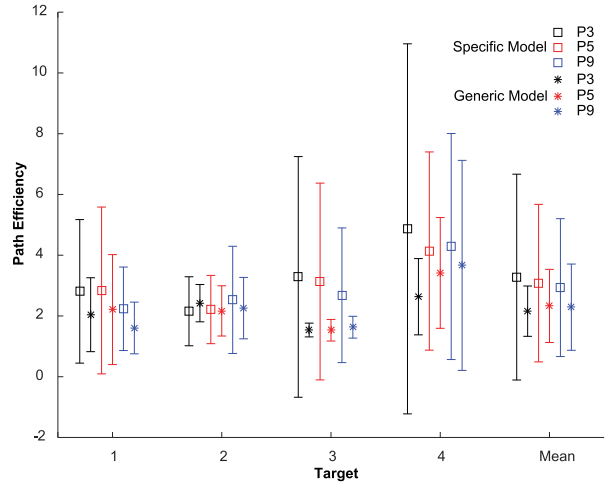

Fig. 7. Average path efficiency for each target across AB subjects and trials. The rightmost six points are the average values across all targets. Error bars represent the standard deviations.

Fig. 7 reports the average path efficiency for each target across AB subjects and trials. The path efficiency was around 3.1 for all targets in the 3 different testing postures for the specific model. The path efficiency was around 2.25 for all targets in the 3 different testing postures for the generic model. The average path efficiency for the specific model in P3, P5, and P9 was 3.28, 3.08, and 2.93, respectively. The average path efficiency for the generic model in P3, P5, and P9 was 2.16, 2.33, and 2.29, respectively. For target 3, the path efficiencies and their standard deviations for the specific model were higher than that for the generic model in the 3 different testing postures. The three-way ANOVA revealed no significant three-way or two-way interactions. ANOVA with only main effects indicated that the path efficiency for the generic model (2.26) was significantly lower ($p < 0.001$) than that for the specific model (3.10). ANOVA with only main effects also indicated that target type had statistically significant effects ($p < 0.001$) on the path efficiency, while posture had no statistical effects (posture: $p = 0.9$) on the path efficiency.

B. Case Study on Amputee Subject

Fig. 8 shows the average completion time for each target across trials in 3 different testing postures for the amputee subject. The average completion time in P3, P5, and P9 was 9.10 s, 11.85 s, and 10.01 s, respectively. Fig. 9 shows the average number of overshoots for each target across trials for

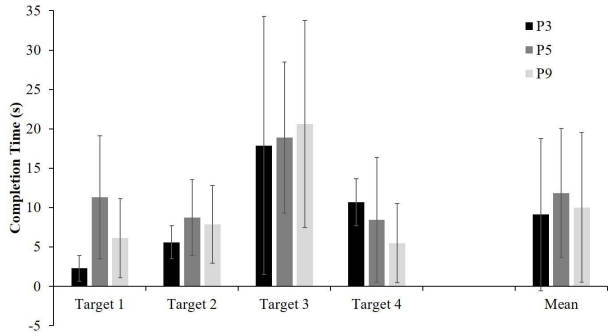

Fig. 8. Average completion time for each target across trials for the amputee subject. The rightmost three bars are the average values across all targets. Error bars represent the standard deviations.

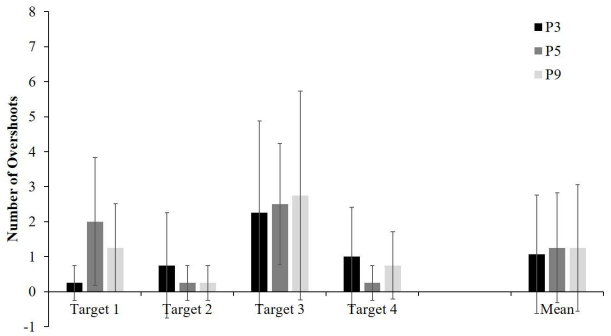

Fig. 9. Average number of overshoots for each target across trials for the amputee subject. The rightmost three bars are the average values across all targets. Error bars represent the standard deviations.

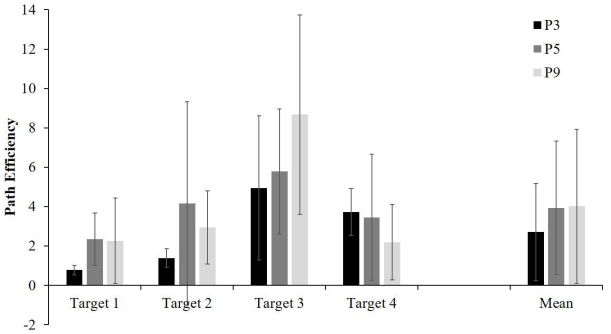

Fig. 10. Average path efficiency for each target across trials for the amputee subject. The rightmost three bars are the average values across all targets. Error bars represent the standard deviations.

the amputee subject. The average number of overshoots in P3, P5, and P9 was 1.07, 1.25, and 1.25, respectively. Fig. 10 shows the average path efficiency for each target across trials for the amputee subject. The average path efficiency in P3, P5, and P9 was 3.65, 5.43, and 5.68, respectively. The amputee accomplished all targets in all trials, even though the three metrics for the amputee were apparently higher than that for the AB subjects.

Fig. 11 shows the input muscle activations applied to the model and output MCP and wrist joint kinematics for a representative trial for target 4 for the amputee subject. The amputee could generate independent and coordinated muscle activations between the MCP and wrist to achieve the target. The muscle activations of flexor or extensor muscles were

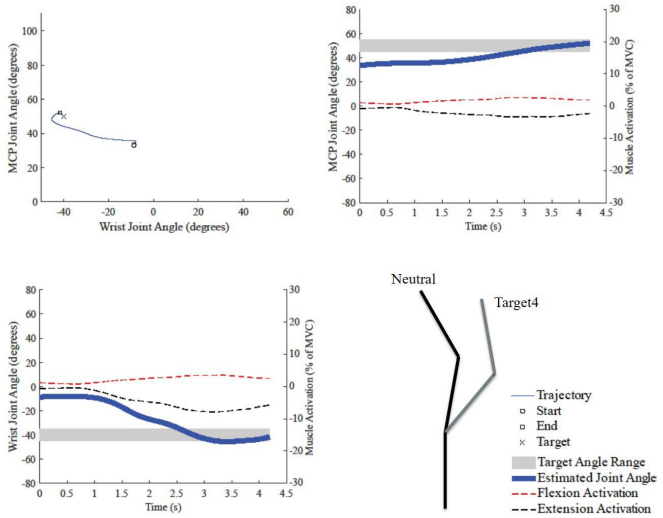

Fig. 11. Input muscle activations applied to the model and output MCP and wrist joint kinematics for the representative trial for target 4 in posture P9 for the amputee subject.

very correlated with the respective flexing or extending joint motions at MCP and wrist joints. For instance, the muscle activations of the wrist flexor muscle were very correlated with changes in the wrist joint angle.

IV. Discussion

In this study we investigated whether one musculoskeletal-model-based EMG NMI can function across different users. We first developed a user generic musculoskeletal model by averaging the model parameters obtained from different AB subjects at different upper limb postures. Impressively, all the subjects recruited in this study, including a transradial amputee, could successfully use the developed NMI to finish the assigned virtual task online after their EMG signals were normalized by MVC. This result indicates that the user-generic musculoskeletal model can capture generalized relationships between neuromuscular control signals and limb motions produced across humans. This generalized relationship was included in the physiology and biomechanical structure/constraints in the neuromusculoskeletal model. Another potential factor that makes our NMI user-generic is human adaptability, which is the user's ability to immediately adjust his or her actions to complete a single real-time control task. During the on-line experiments in this study, human subjects were adaptive and could modify EMG activation level volitionally to produce the desired magnitude of joint motion to achieve the target hand postures. It would be interesting further research to systematically explore human adaptability while using a musculoskeletal model-based EMG NMI. In summary, this study showed that it is possible to develop an EMG NMI that is generalizable across users, and our developed musculoskeletal-model-based EMG NMI can potentially serve as a multi-user NMI for various applications in the future, including myoelectric prosthesis.

In contrast to the existing data-driven NMI, the developed generic musculoskeletal model showed reliability across both users and postures and simplified and shortened the

calibration process. Such properties might relieve the burden of frequent and cumbersome recalibrations, usually needed for the existing data-driven EMG NMI. However, the success of our generic model requires muscle specific recordings, while data-driven approaches are usually flexible with the EMG electrode placement. This requirement brings some challenge for upper limb amputees; imaging approaches (such as ultrasound) are desired to identify the exact location of residual muscles (if they exist) in upper limb amputees in order to identify appropriate locations for EMG electrode placement. For amputees who have undergone targeted muscle reinnervation surgeries [34], [35], additional surgical or engineering techniques are needed to isolate function specific muscle activity before applying our generic model-based NMI for prosthesis control.

It was a very interesting observation that the generic model outperformed the specific model with a reduced number of overshoots and improved path efficiency. We initially hypothesized that the performance of the user-specific model should be better than that of the generic model since the specific model was tailored to each individual subject. However, our on-line experimental results demonstrated that compared to the specific model, the generic model produced comparable task completion times, significantly reduced the number of overshoots, and significantly improved path efficiency. One possible explanation was that the on-line experiments were conducted on a different day, and EMG electrodes were not placed at the exact same locations each day. Since the generic model was built by data collected from multiple users at multiple postures, it could be more generalized to the experiment setup and less sensitive to the electrode placement and other EMG interface variations, compared to the user-specific models. Consequently, the generic model yielded slightly better on-line control performance in the model-based EMG interface.

Impressively, the amputee subject recruited in this study completed all the trials using the generic model built by the data collected form AB subjects. It is noteworthy that this amputee had been only exposed to our EMG-driven generic musculoskeletal model two times (about 4 hours practice) before the online testing experiment. This observation might indicate that the internal representation of the musculoskeletal system of the amputated limb somehow still partially exists. We targeted the muscles using surface EMG electrodes through careful palpation on the residual limb to ensure independent muscle activities as the musculoskeletal model input. Therefore, the generic musculoskeletal model developed from the AB subjects could still work for this transradial amputee. The reduced performance in the amputee subject, compared to AB subjects, was potentially due to neural plasticity after limb loss [36]. Since the amputee subject was only introduced to the musculoskeletal model-based EMG NMI for a limited amount of time, perhaps more training might further improve the on-line control performance of this amputee subject. This is one of our future studies.

The present study was the initial step to show that it is feasible to design a user-generic musculoskeletal model-based EMG NMI. Since only six AB subjects' data were acquired in the off-line experiment to develop the user-generic model, it will be an interesting future study to explore other more sophisticated methods to extract user-generic parameters when we collect more data from more human subjects. Since the development of the model-based control approach is still at an early stage, we did not compare it to a more established (either direct control or pattern recognition) technique, which is a limitation of the current study. A comparison to more established techniques should be conducted in the future. Our long-term goal is to make our novel NMI an alternative control option for myoelectric prostheses. Since every control approach has its pros and cons, we intend to make an appropriate combination of these two control approaches to leverage each of their advantages for more effective control in the future. The results derived from testing of one amputee is encouraging. However, to achieve our long-term goal, it is our future work to test more amputees and evaluate the model-based NMI in functional task performance. Another limitation of our study was that we could not separate the contributions from the human adaptability and the algorithm during the on-line control test. Human adaptation is a critical factor in the performance of a human-machine integrated system. Therefore, a systematic study of human adaptability while using the model-based EMG NMI should be performed in the future. The user-generic model could be potentially generalized not only to across-user and across-posture variations as shown in the current study, but also to other sources of variations (i.e. across-electrode-shift variations), which should be systematically evaluated in our future study. The research also elicited an interesting question: when user-generic NMI is available, is it necessary to personalize the EMG NMI specifically to individual users? Perhaps the generic NMI provides a good initial design, and the best practice for personalization should be to slightly adjust the generic model parameters for each user. We will explore this idea in the future. Finally, two DoFs were considered in the generic musculoskeletal model in this study. Since hand rotators have been commercially available for upper limb amputees, we added wrist supination/pronation into the musculoskeletal model [33]. We plan to add this additional function into the generic NMI in our next study phase.

V. Conclusions

In this study, we presented a novel EMG NMI based on a user-generic musculoskeletal model. The EMG interface was tested on seven AB subjects and one amputee subject while performing virtual hand posture matching experiments. Both the AB subjects and the amputee subject completed all the assigned virtual tasks, although the performance of the amputee subject was poorer than that of the AB subjects. Compared with the specific model, the generic model produced comparable completion time, reduced the number of overshoots, and improved path efficiency in the virtual task performance. The results suggested that it was possible to use a generic musculoskeletal model to continuously decode EMG signals for predicting MCP and wrist joint angles for both the AB subjects and the amputee subject. Our future study will focus on evaluating the developed NMI for powered prosthetic arms.

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